

* Non-rel. QM of a single Particle in 1d

$$H(P, Q) = \frac{1}{2m} P^2 + V(Q)$$

$$\{Q, P\} = i \quad (\hbar = 1)$$

• We want to evaluate the probability amplitude of going from $(t', q') \rightarrow (t'', q'')$

$$\langle q'' | e^{-iH(t''-t')} | q' \rangle$$

Heisenberg Pic: $Q(t) = e^{iHt} Q(0) e^{-iHt}$

$$|q(t)\rangle = e^{iHt} |q\rangle$$

$$\langle q'', t'' | q', t' \rangle$$

• Time interval: $T \equiv t'' - t'$

Break into $(N+1)$ equal pieces

$$\delta t = T/(N+1)$$

$$e^{-iHT} = e^{-iH(N+1)\delta t} = \overbrace{e^{-iH\delta t} \dots e^{-iH\delta t}}^{N+1}$$

• Completeness relation

$$1 = \int dx_i |x_i\rangle \langle x_i|$$

Introduce N -complete sets

$$\langle q'', t'' | q', t' \rangle = \left(\prod_{i=1}^N \int dx_i \right) \langle q'' | e^{-iH\delta t} | q_N \rangle \langle q_N | e^{-iH\delta t} | q_{N-1} \rangle \dots \langle q_1 | e^{-iH\delta t} | q' \rangle$$

• Since $H = \frac{1}{2m} \tilde{p}^2 + V(\tilde{q})$

$$\langle q_i | e^{-i(\frac{1}{2m}\tilde{p}^2 + V(\tilde{q}))\delta t} | q_j \rangle$$

$$\Rightarrow \exp(-\frac{p^2}{2m}\delta t) \exp(-iV(q)\delta t) \exp(i\mathcal{O}(\delta t^2))$$

take $N \rightarrow \infty$ so last term is vanishing

$$= \langle q_i | e^{-i\frac{\delta t}{2m}\tilde{p}^2} e^{-iV(\tilde{q})\delta t} | q_j \rangle$$

$$= \int dp_i \langle q_j | p_i \rangle e^{-i\frac{\delta t}{2m}p_i^2} e^{-iV(q_i)\delta t} \langle p_i | q_i \rangle$$

$$|\psi\rangle = \int d^d x \langle x|\psi\rangle |x\rangle$$

$$\langle p|\psi\rangle = \int d^d x \langle x|\psi\rangle \langle p|x\rangle$$

$$\psi(p) = \int d^d x e^{-i p x} \psi(x)$$

$$\langle p|x\rangle = e^{-i p x} / \sqrt{2\pi}$$

$$\langle x|p\rangle = e^{i p x} / \sqrt{2\pi}$$

$$\begin{aligned} &= \int dp \left(e^{i q_1 p} / \sqrt{2\pi} \right) e^{-i \frac{\delta t}{2m} p^2} e^{-i V(q_1) \delta t} e^{-i p q_1} / \sqrt{2\pi} \\ &= \int (dp / 2\pi) e^{-i H(p, q_1) \delta t} \exp(i p_1 (q_2 - q_1)) \end{aligned}$$

$$\langle q_2 | e^{-i H \delta t} | q_1 \rangle = \int \frac{dp}{(2\pi)} e^{-i H(p, q_1) \delta t} e^{i p_1 (q_2 - q_1)}$$

* Adopt Weyl ordering

$$\hat{H}(p, q) \equiv \int \frac{dx dq}{(2\pi)^2} e^{i x \hat{p} + i q \hat{q}} \int dp dq e^{-i x p - i q q} f(p, q)$$